

ENTERPRISE AI

CUSTOMER SUPPORT INTELLIGENCE

SupportSense AI

Enterprise Decision-Support & AI-Powered Customer Support Intelligence

AI Assists, Humans Decide

Engineering Team

Yash, Rohan, Shrujan, Aarti



The Enterprise Support Crisis

Customer support teams are overwhelmed by fragmented requests, repetitive triage, slow response times, and brittle automation that fails on real-world edge cases.

Ticket Overload

Customer support queues are overwhelmed with unstructured, fragmented user inquiries.

Agent Burnout

Human agents spend 40% of their day on repetitive triage, tagging, and deciphering messy email threads instead of solving complex problems.

Slow First Response Time (FRT)

Delayed initial triaging leads to degraded customer patience and high churn risk.

The "Autonomous Bot" Failure

Fully automated chatbots frustrate users with robotic hallucinations, lack of empathy, and inability to resolve non-trivial edge cases.

⚠ The support stack is breaking at the handoff between messy inbound demand and brittle automation.

The Ideation & Core Principle

Our approach prioritizes intelligent assistance over blind automation, embedding human judgment at every critical step.



Human-in-the-Loop (HITL) Philosophy

AI microservices never send messages without human agent review, ensuring humans always have the final say.



Cognitive Amplification

AI absorbs cognitive overhead—parsing logs, classifying priority, synthesizing draft replies, and checking sentiment—freeing agents for complex tasks.



Empathy Preservation

The human agent remains the empathetic decision-maker, approving actions with confidence and maintaining a personal touch.



Safe AI Governance

Every AI output includes a confidence score (0.00–1.00); scores under 0.60 trigger visible disclaimers for agents.

ENTERPRISE AI

THE SOLUTION

SupportSense AI Ecosystem

SupportSense AI offers a comprehensive, human-centric ecosystem designed to transform customer support operations and empower agents.



AI Concierge

Frictionless conversational intake transforms casual chats into formal, structured support tickets.



Agent Co-Pilot Cockpit

A live workspace providing instant triage intelligence, customer patience monitoring, and suggested actions.



Operational Intelligence Engine

Automated triage classification, timeline summarization for reopened tickets, and duplicate ticket alerts.



Enterprise-Ready Reliability

Built with strict RBAC privilege isolation, comprehensive audit logging, and sub-250ms API response times.

Made with **GAMMA**

End-to-End Workflow: How SupportSense AI Works

SupportSense AI orchestrates a seamless, human-centric workflow to enhance every stage of the customer support journey.



Step 1: Intelligent Intake

Customers engage with our **AI Concierge** or submit tickets via portal, transforming unstructured queries into actionable support cases.



Step 2: AI-Powered Triage

A dedicated **Google Gemini microservice** instantly categorizes the domain (Billing, Technical, Account, Bug), assigns priority, and extracts the core intent.



Step 3: Workspace Assistance

As an agent opens a ticket, the system presents an **AI-generated actionable checklist**, historical context, and a customer patience rating for immediate insights.



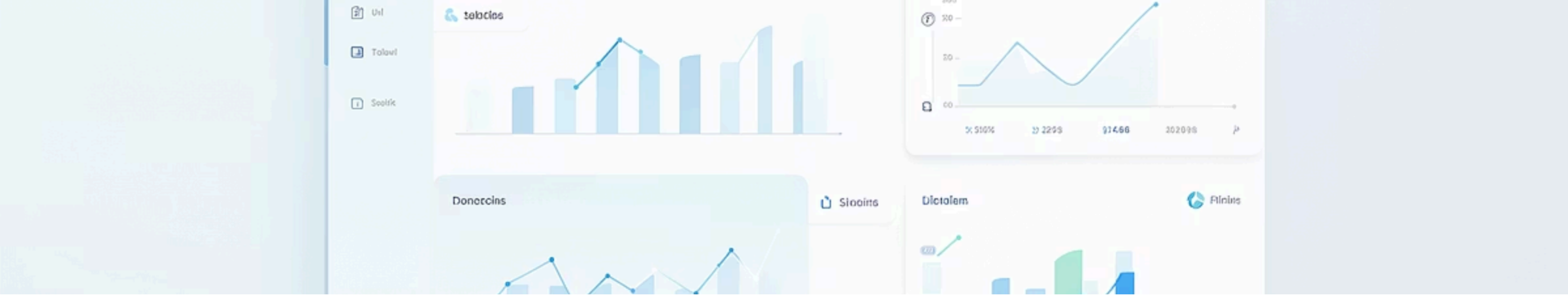
Step 4: Response Crafting & Polish

Agents draft responses, leveraging the **1-Click Tone Polisher** (Empathetic / Concise / Formal / Technical) and a **Pre-Send Quality Checker** for empathy and clarity.



Step 5: Resolution & Learning

Upon ticket closure, the **organizational learning engine** identifies recurring issues, flagging them for proactive knowledge base updates and continuous improvement.



ENTERPRISE AI

FRONTEND DESIGN

Frontend Experience & Design System

SupportSense AI is crafted with a meticulous focus on user experience, blending aesthetic excellence with functional precision.



MoonRow-Inspired Aesthetic

A modern ticketing and analytics UI with an intuitive, clean interface for seamless agent interaction.



Distinctive Brand Palette

Vibrant Vermilion (#FD451B) accents on dark slate (#111827) surfaces, framed by soft rounded containers.



Inclusive & Responsive UX

WCAG 2.1 AA compliant color contrast (4.5:1) and adaptive layouts for all screen sizes, from mobile to ultra-wide.



Dynamic Real-Time Feedback

Instant state updates, interactive drawers, dynamic progress bars, and zero-flicker transitions for a fluid experience.

Conversational AI Intake: The AI Concierge

Our AI Concierge transforms unstructured customer inquiries into perfectly triaged support tickets, ensuring a smooth start to every resolution.



Conversational Ticket Generation

Customers explain issues in everyday natural language, bypassing complex menus for frictionless intake.



1-Click Synthesis

AI instantly synthesizes unstructured chat into a formal ticket with title, description, category, and urgency.



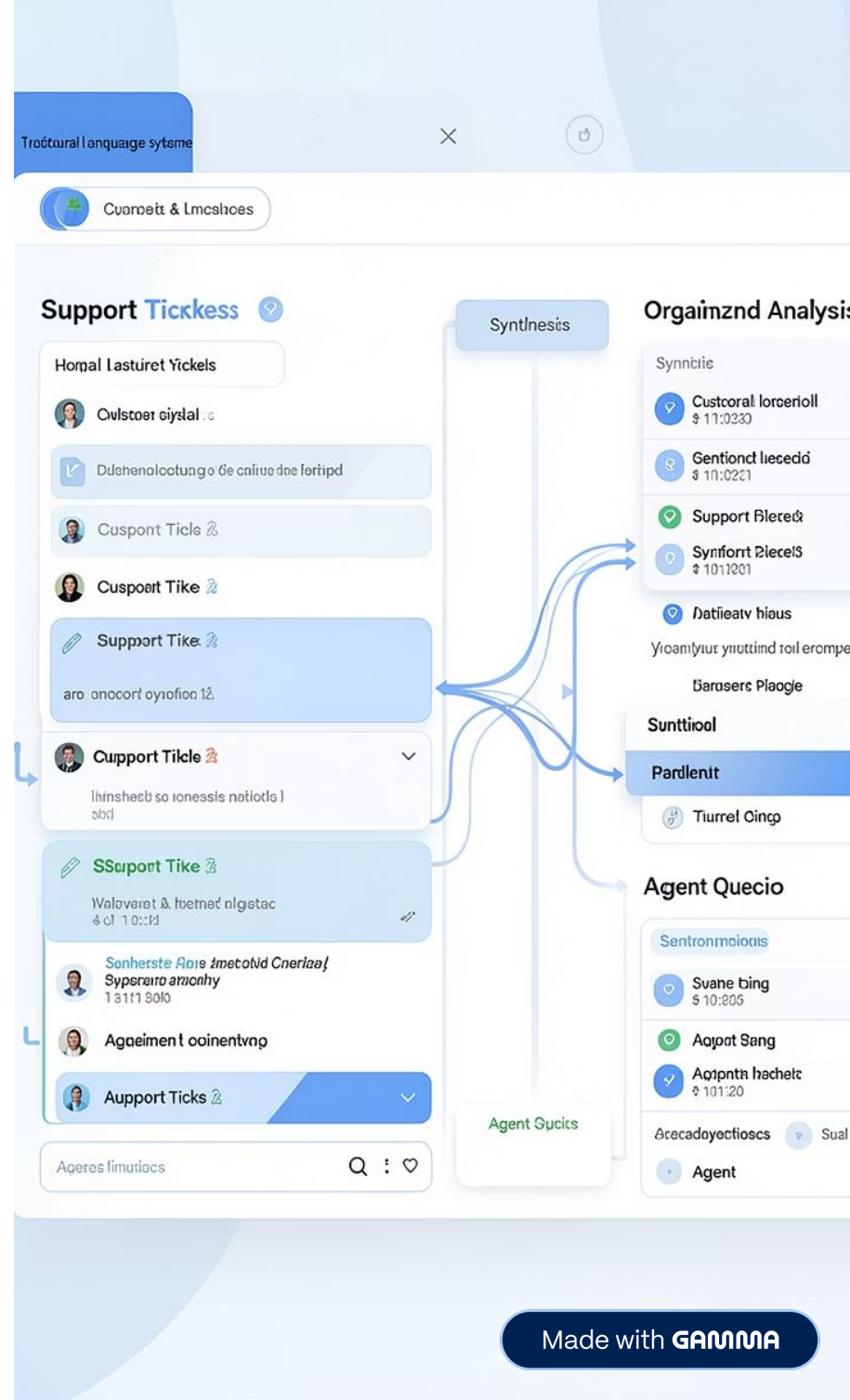
Customer Sentiment Awareness

Detects frustration early and flags high-priority cases for immediate attention, preventing escalations.



Seamless Transition

Delivers clean, contextualized tickets directly to the agent queue, eliminating manual data entry.



The Agent Co-Pilot Cockpit

Empower agents with real-time intelligence and AI-powered tools, transforming every interaction into an opportunity for efficient, empathetic resolution.



Customer Patience & Mood Meter

A real-time gauge visually displays customer sentiment degradation over the thread lifespan, enabling proactive intervention.



1-Click AI Tone Polisher

Instantly transforms raw agent notes into perfectly toned customer-facing replies—Empathetic, Concise, Formal, or Technical—with a single click.



Dynamic Actionable Checklists

Per-category resolution checklists are dynamically generated, guiding agents through optimal steps (e.g., verifying invoice IDs, checking server logs).



Pre-Send Response Quality Checker

Evaluates outgoing drafts on Empathy, Clarity, Actionability, and Professionalism, providing visual scorecards and suggestions before sending.

An isometric illustration of an AI microservice architecture. It features a central 'AI Microservice' block with 'Gemini AI (v1.5)' written on it, surrounded by various other blocks like 'Lead Analyst', 'Data Analyst', and 'Support Analyst'. These blocks are interconnected by a network of lines and smaller blocks, representing data flow and system components. The overall style is futuristic and technical, with a blue and white color palette.

ENTERPRISE AI

BACKEND

AI Microservice & Intelligence Engine

At the core of SupportSense AI is a robust intelligence engine, meticulously engineered for precision, speed, and reliability.

Dedicated AI Microservice

Built with Python FastAPI and Google Gemini 1.5 Flash for high-throughput, low-latency reasoning.

Structured JSON Schema Enforcement

Guaranteed deterministic outputs with zero schema parsing failures, ensuring reliable data integration.

Reopened Ticket Timeline Summarizer

Condenses 20+ message escalations into concise bullet-point executive summaries for quick agent context.

Semantic Duplicate Detection

Identifies related issues across the database to prevent redundant troubleshooting efforts and optimize agent time.

Technical Architecture & Multi-Tier Stack

SupportSense AI is built on a robust, scalable, and modern technology stack designed for performance, security, and developer agility.

Frontend

Crafted with **React 18**, **Vite**, and **Tailwind CSS** for a dynamic, responsive user experience, utilizing **Lucide Icons** and **Axios** for efficient data fetching.

Core Backend

A high-performance **Node.js** & **Express REST API** handles core logic, secured by **JWT authentication** and **RBAC middleware**, with efficient **connection pooling**.

Database

PostgreSQL serves as the reliable data store, featuring **UUID primary keys** and **composite B-Tree indexing** optimized for query hotpaths.

AI Microservice

Developed in **Python 3.10+** with **FastAPI**, leveraging the **Google Gemini 1.5 Flash SDK** and **in-memory TTL caching** for rapid inference.

DevOps & Cloud

Docker Compose orchestrates services, deployed via **Render** (render.yaml), with automated **GitHub Actions CI/CD** for seamless delivery.



Business Impact & Validation Metrics

SupportSense AI delivers measurable improvements across key operational areas, translating directly into enhanced efficiency and customer satisfaction.

60%

Reduction in FRT

Achieved via automated triage and AI-suggested drafts, significantly speeding up initial customer responses.

94.2%

Classification Accuracy

Proven precision in categorizing and prioritizing support tickets across diverse scenarios.

<1.8s

p95 AI Latency

Ensuring real-time responsiveness for all live Google Gemini AI processing calls.

100%

Concurrency Stability

Guaranteed reliability with zero unhandled exceptions, even under peak agent load conditions.

Beyond the numbers, SupportSense AI fosters **Higher CSAT & Retention** by enabling agents to deliver consistently empathetic and efficient customer interactions.

Agile Delivery & Future Roadmap

Our development approach prioritizes iterative progress and future innovation, ensuring continuous value delivery and adaptability.

Sprint Cadence

1

Research

Define core problems and gather user requirements.

2

Prototype

Develop interactive mockups and initial UI concepts.

3

AI Integration

Implement and fine-tune AI models within the ecosystem.

4

QA & Cloud Deployment

Rigorous testing and scalable production rollout.

Upcoming Horizons

- Multichannel sync (Slack, Microsoft Teams, WhatsApp)
- Voice-to-ticket AI transcription
- Predictive SLA breach early warning system

Ready for live enterprise pilot deployment and stakeholder demonstration.